Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=5;i++)
4       (i-1)[arr] = (i%5);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 3;
8   }
9  int main(){
10   int arr[6] = {2,2,2,2,2};
11   func1(arr+1);
12   for(int i=0; i<6; i++)
13         printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<6; i++)
17         printf("%d ", i[arr] );
18   return 0;
19 }
```

a[0] = arr+3

arr[0] = a[1]

array of one element

arr+3;

{2,2,2,2,2,0}

{2,1,2,3,4,5}

1 5 4 3 2

1 5 4 3 2

0,1,2,3,4,5

pointer to arr

arr → *arr + 3

Output
2 1 2 3 4 5
2 2 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 4;
 a = &temp;
}
```

Explain your answer

The answer would change because now the address of 0th element changes, hence the array begins from a new position as compared to before.

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

50 50

→ Address of arr1[0] & arr2[0] is stored by arr1 & arr2 respectively, and the following elements' addresses &(arr1[n]) = arr1 + n, regardless of whether the array contains integers or long integers.